

# Workshop - Week 1

1. (a) Solve the equations and inequalities for  $x$ 
  - (1)  $|2x| = 3$
  - (2)  $|x + 3| = |2x + 1| \leftrightarrow x + 3 = 2x + 1$  or  $x + 3 = -(2x + 1)$ .
  - (4)  $|x| < 3$
  - (5)  $|x - 4| < 1$

(b) *Harder exercise:* Using  $|a| = \sqrt{a^2}$  prove that  $|ab| = |a||b|$

(c) *Harder exercise:* Show that the sum, difference, and product of rational numbers are rational numbers.
2. (a) Find the slope of the line passing through  $(1, 5)$  and  $(4, 11)$ 

(b) Find the equation of the line passing through  $(2, 1)$  with slope 6

(c) Find the slope and  $y$ -intercept of the line and draw its graph

  - (1)  $x + 3y = 0$
  - (2)  $3x - 4y = 12$

(d) Sketch the regions in the  $xy$ -plane

  - (1)  $\{(x, y) \mid x < 0\}$
  - (2)  $\{(x, y) \mid xy < 0\}$
  - (3)  $\{(x, y) \mid |x| < 2\}$
3. (a) Show that the equation  $x^2 + y^2 - 4x + 10y + 13 = 0$  represents a circle and find the center and radius

(b) Identify the type of curves

  - (1)  $16x^2 - 25y^2 = 400$
  - (2)  $y = x^2 - 6x + 13$

(c) Sketch the graph of the set  $\{(x, y) \mid y > x^2 - 1\}$
4. (a) Prove that:
  - (1)  $\cos(\frac{\pi}{2} - x) = \sin(x)$
  - (2)  $\sin(\frac{\pi}{2} - x) = \cos(x)$
  - (3)  $\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$
  - (4)  $\sin(x) \sin(2x) + \cos(x) \cos(2x) = \cos(x)$

(a) Find the values of  $x$  in the interval  $[0, 2\pi]$  that satisfy the equation

  - (1)  $2 \sin^2(x) = 1$
  - (2)  $\sin(2x) = \cos(x)$